TASK-1 Business Sales Performance

In [1]:
```
import pandas as pd
import matplotlib.pyplot as plt
```

In [2]:
```
df = pd.read_csv("D:\sales_data.csv")
df.head()
```

Out[2]:

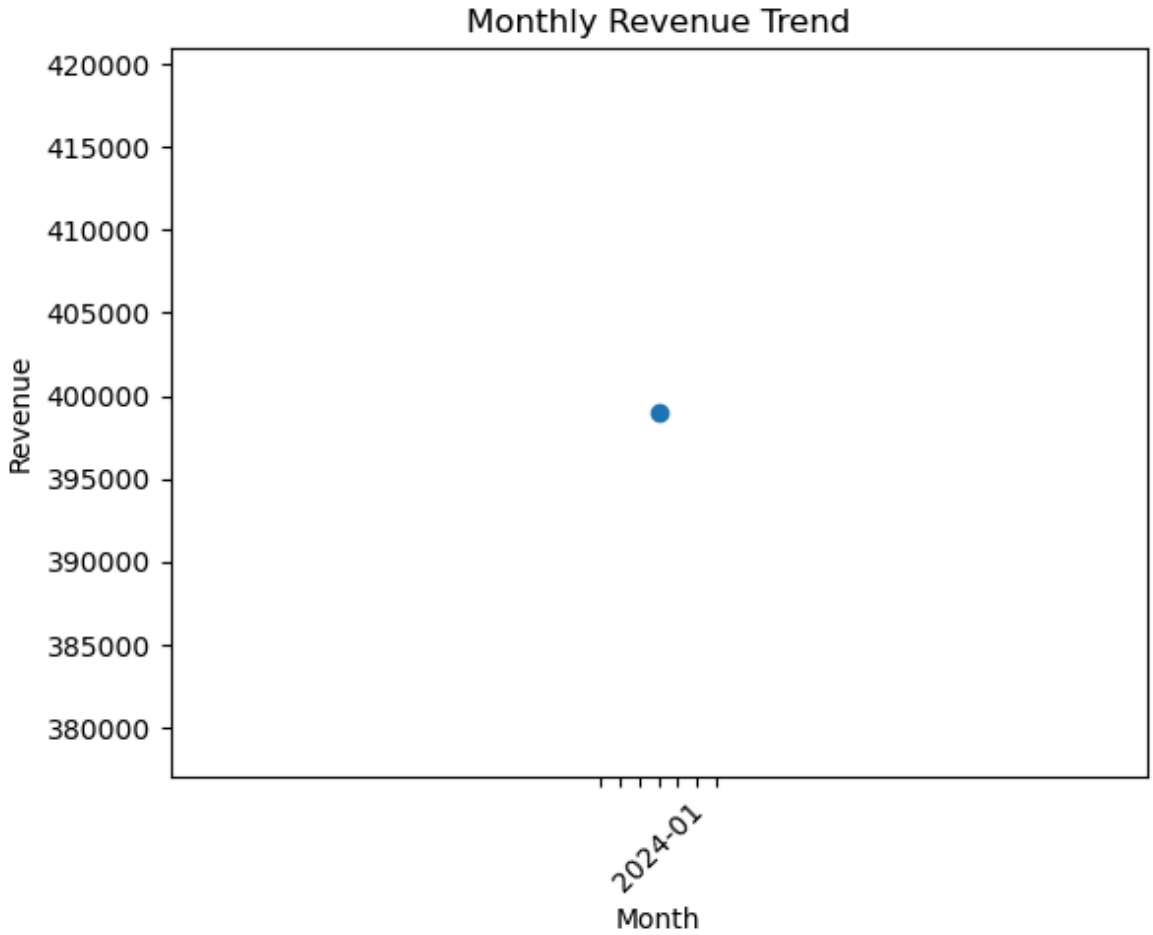
	Customer_ID	Age	Gender	City	Product	Price	Quantity	Purchase_Date
0	101	23	Male	Bangalore	Laptop	55000	1	2024-01-10
1	102	30	Female	Mumbai	Phone	20000	2	2024-01-11
2	103	22	Female	Delhi	Headphones	2000	3	2024-01-12
3	104	35	Male	Bangalore	Laptop	55000	1	2024-01-15
4	105	28	Female	Chennai	Phone	20000	1	2024-01-18

In [3]:
```
df.drop_duplicates(inplace=True)
df.dropna(inplace=True)

# Convert date column
df['Purchase_Date'] = pd.to_datetime(df['Purchase_Date'])

# Create Month (as string for better plotting)
df['Month'] = df['Purchase_Date'].dt.strftime('%Y-%m')
```

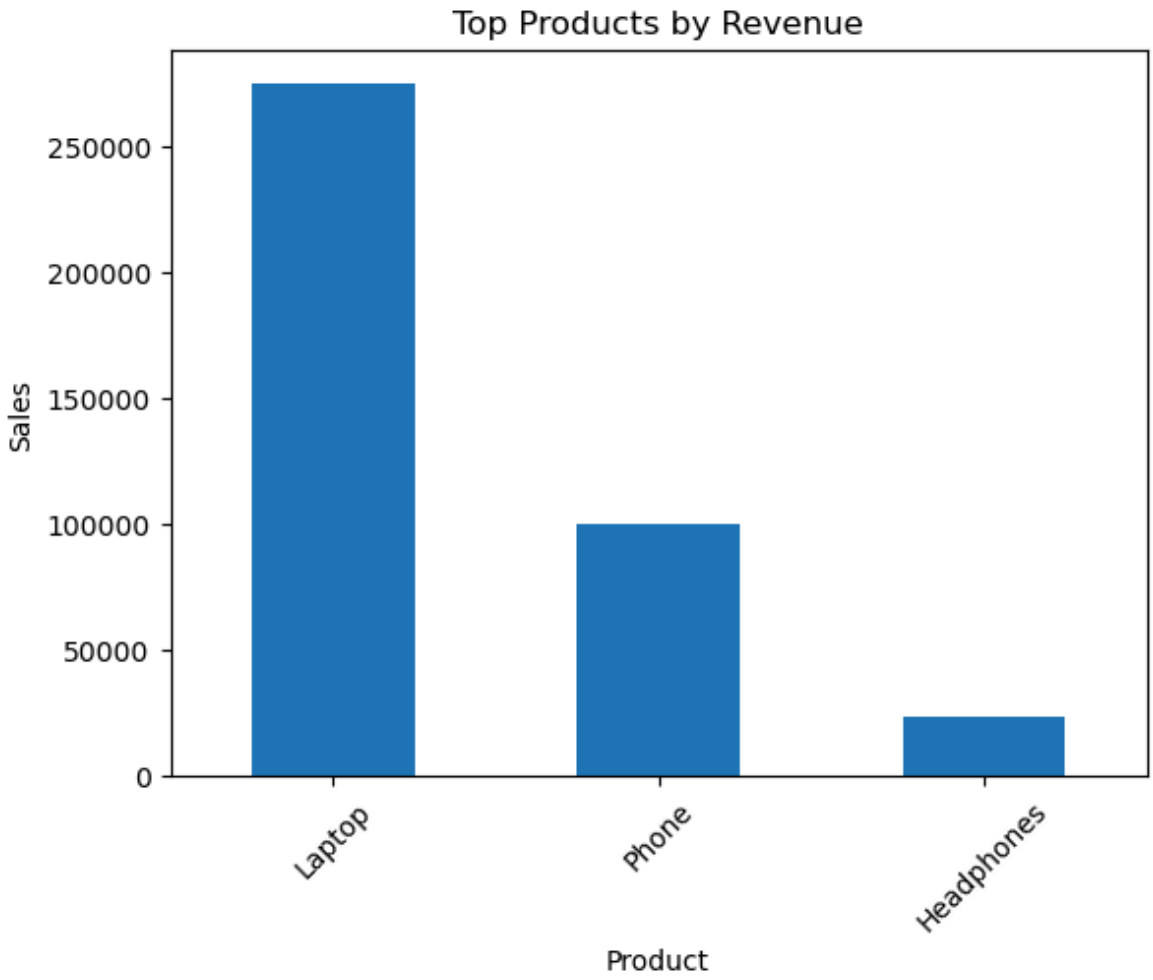
In [4]:
```
df['Sales'] = df['Price'] * df['Quantity']
```

In [5]:
```
# Group
trend = df.groupby('Month')['Sales'].sum()

print(trend)  # <-- helps confirm data

# Plot (force visible markers)
trend.plot(marker='o')

plt.title("Monthly Revenue Trend")
plt.xlabel("Month")
plt.ylabel("Revenue")
plt.xticks(rotation=45)

# Fix for single data point
if len(trend) == 1:
    plt.xlim(-0.5, 0.5)

plt.show()
```

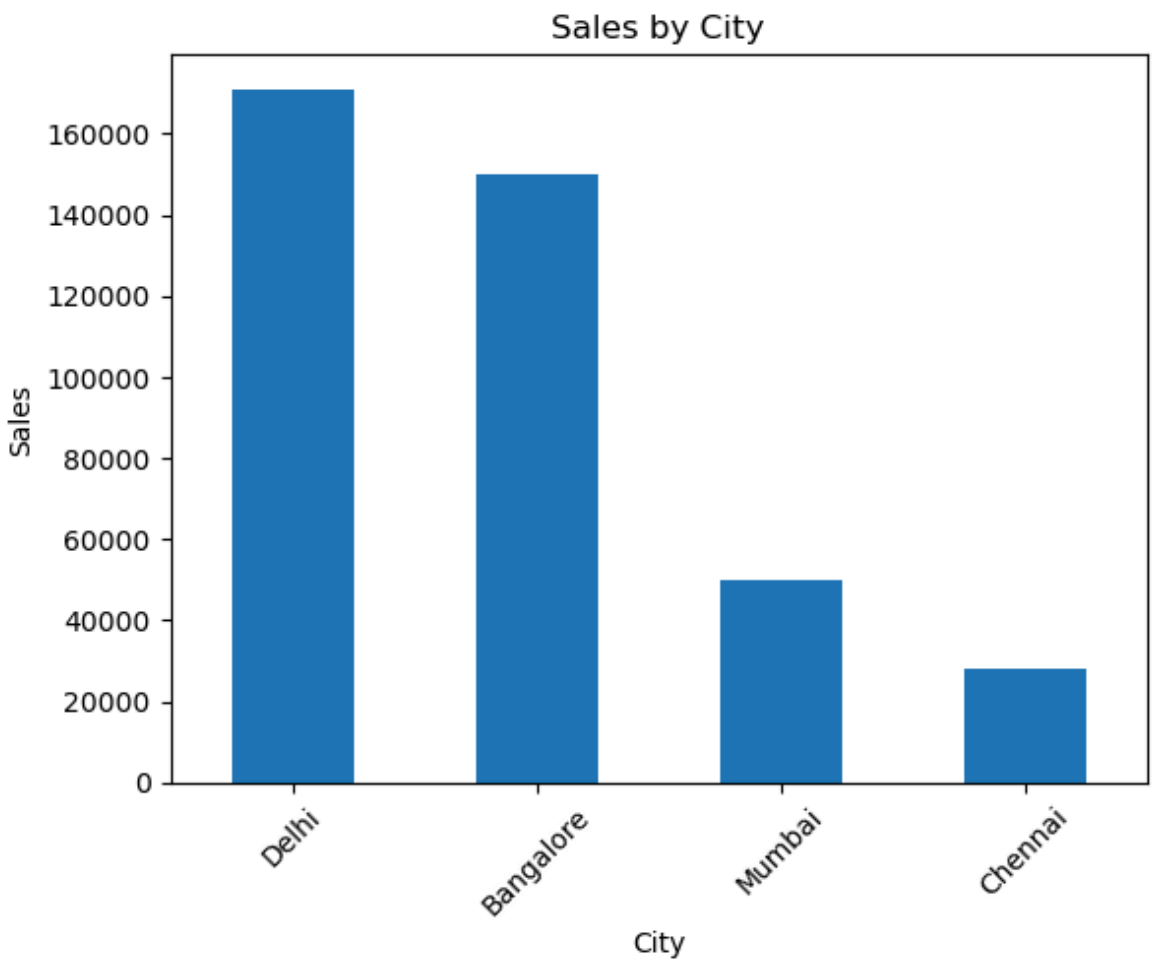
```
Month
2024-01    399000
Name: Sales, dtype: int64
```



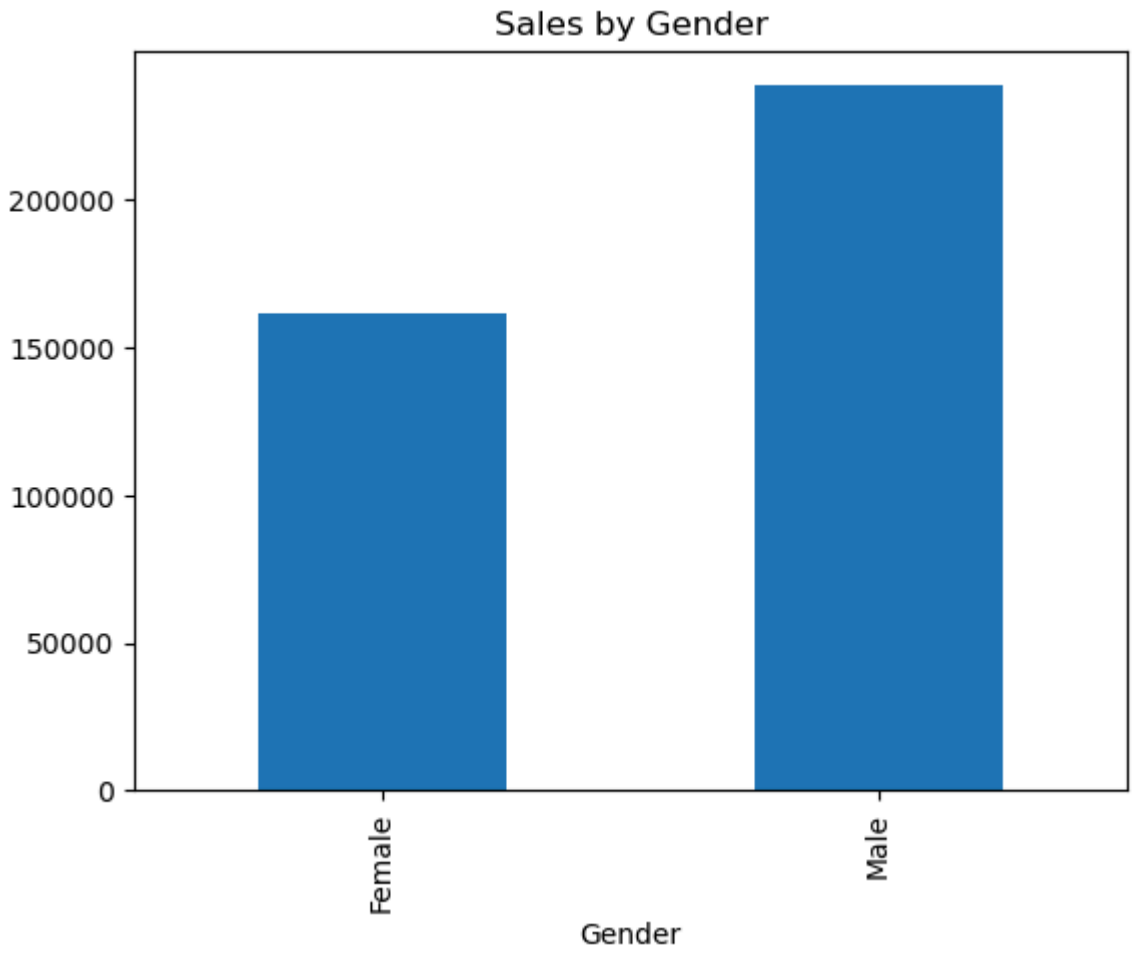
In [6]:
```
top_products = df.groupby('Product')['Sales'].sum().sort_values(ascending=False)

top_products.plot(kind='bar')
plt.title("Top Products by Revenue")
plt.xlabel("Product")
plt.ylabel("Sales")
plt.xticks(rotation=45)
plt.show()
```



In [7]:
```
city_sales = df.groupby('City')['Sales'].sum().sort_values(ascending=False)

city_sales.plot(kind='bar')
plt.title("Sales by City")
plt.xlabel("City")
plt.ylabel("Sales")
plt.xticks(rotation=45)
plt.show()
```



In [8]:
```
gender_sales = df.groupby('Gender')['Sales'].sum()

gender_sales.plot(kind='bar')
plt.title("Sales by Gender")
plt.show()
```



Key Insights

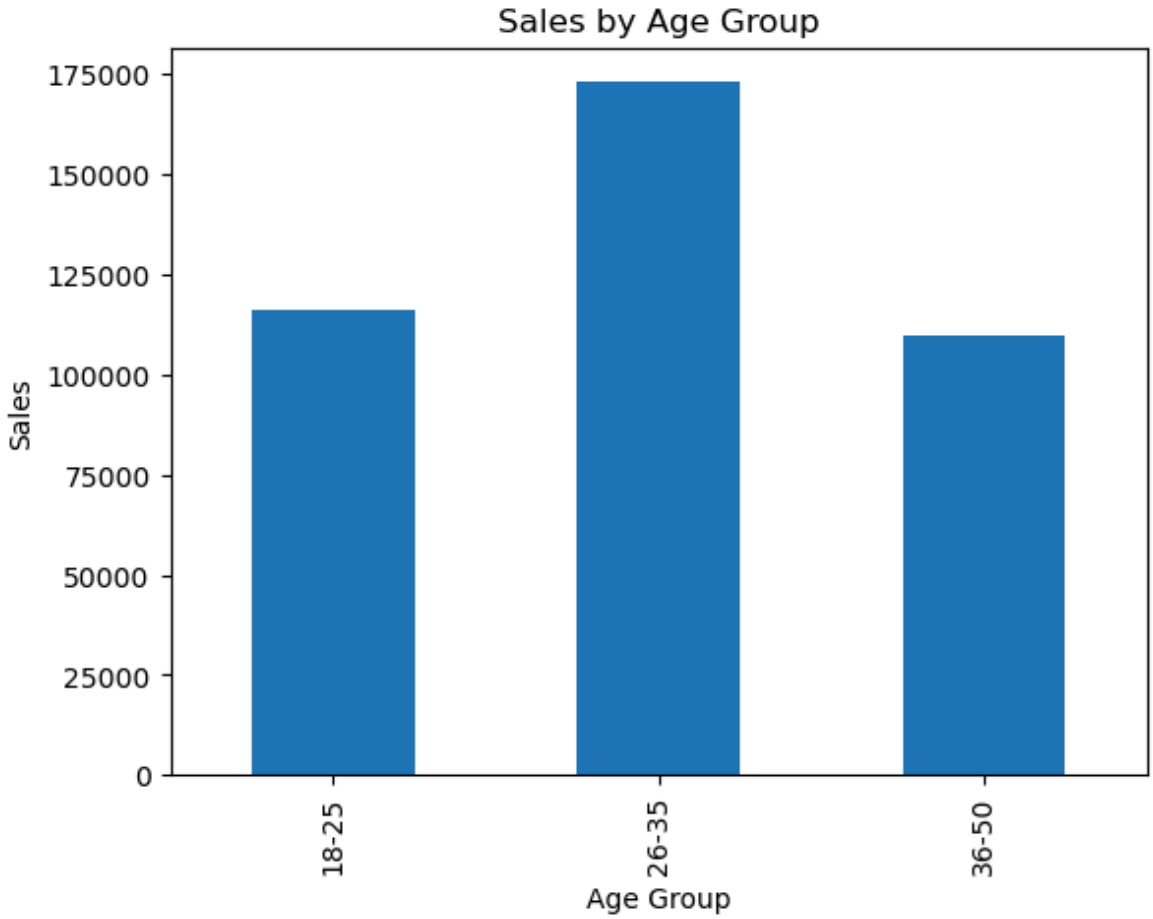
- The highest revenue is generated by [Top Product]
- [City Name] contributes the most to total sales
- Sales show a [increasing/decreasing/fluctuating] trend over time
- [Male/Female] customers contribute more to revenue
- Some products have significantly lower sales indicating low demand

TASK-2 Customer analysis

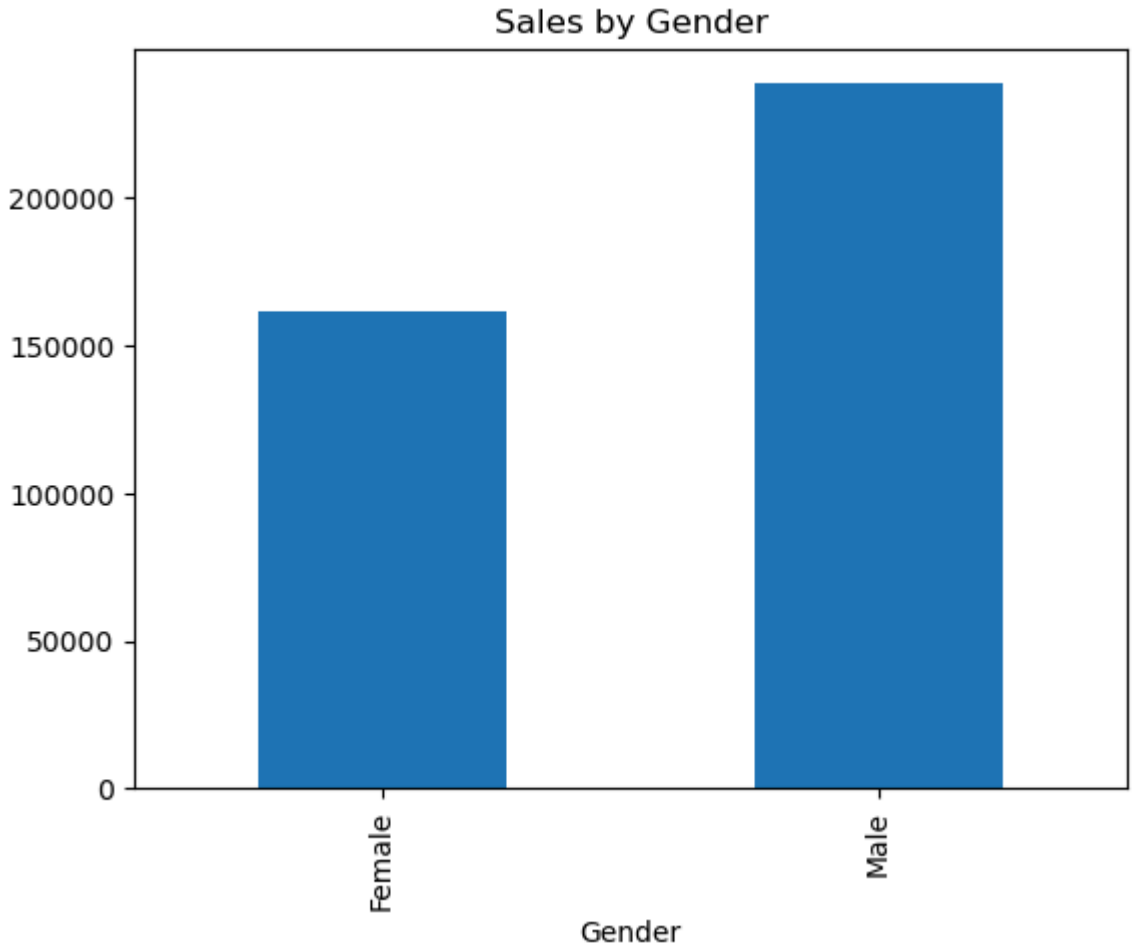
In [9]:
```
df['Age_Group'] = pd.cut(df['Age'],
                         bins=[18,25,35,50],
                         labels=['18-25','26-35','36-50'])
```

In [10]:
```
age_sales = df.groupby('Age_Group')['Sales'].sum()

age_sales.plot(kind='bar')
plt.title("Sales by Age Group")
plt.xlabel("Age Group")
plt.ylabel("Sales")
plt.show()
```

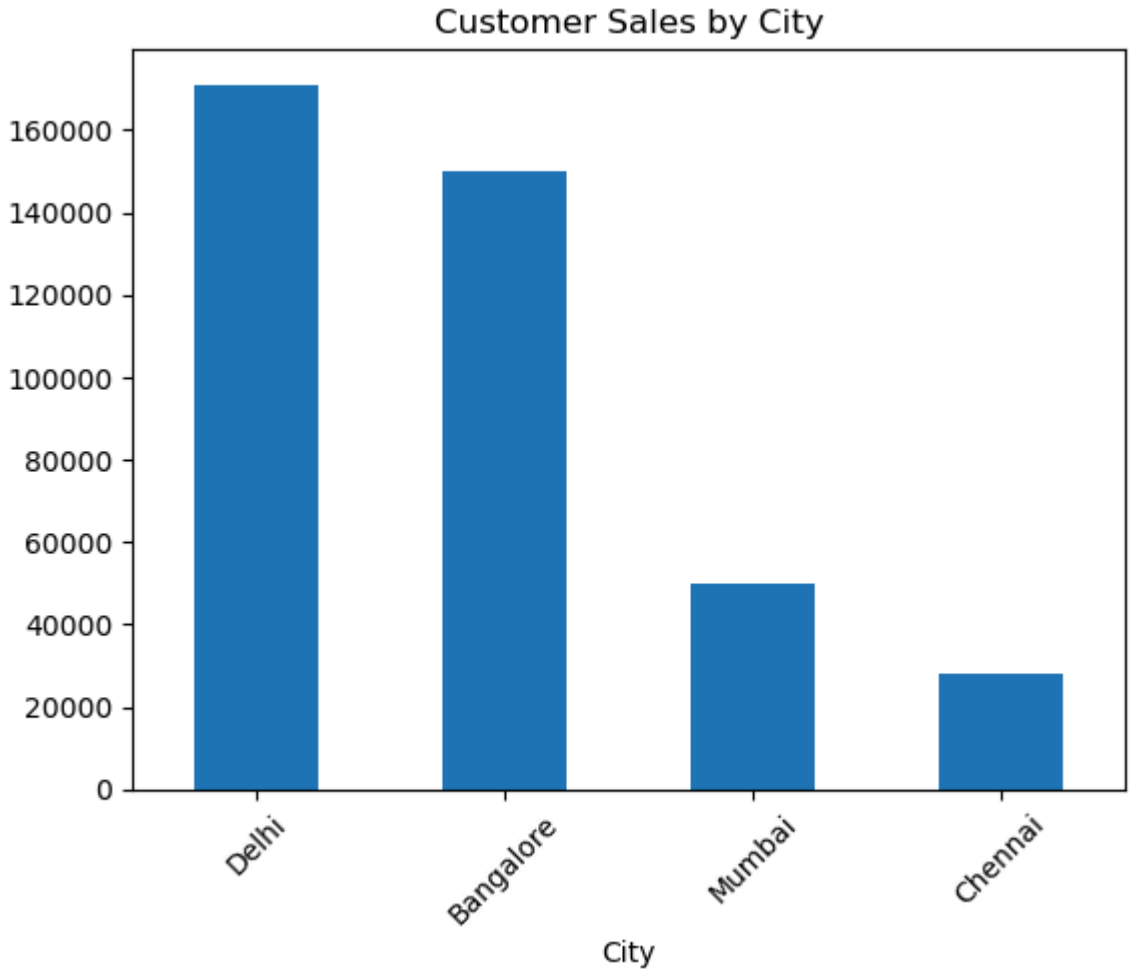
```
C:\Users\Admin\AppData\Local\Temp\ipykernel_2796\3041588383.py:1: FutureWarning:
The default of observed=False is deprecated and will be changed to True in a futu
re version of pandas. Pass observed=False to retain current behavior or observed=
True to adopt the future default and silence this warning.
  age_sales = df.groupby('Age_Group')['Sales'].sum()
```



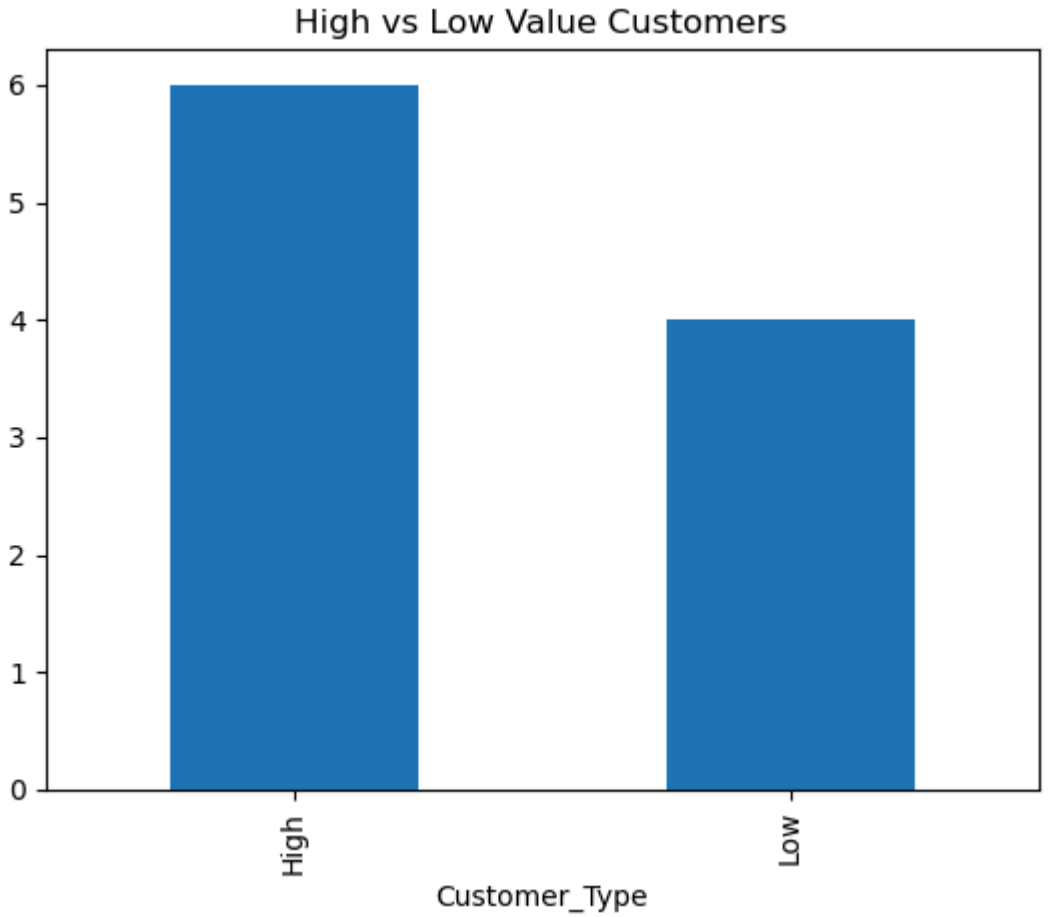
In [11]:
```
gender_sales = df.groupby('Gender')['Sales'].sum()

gender_sales.plot(kind='bar')
plt.title("Sales by Gender")
plt.show()
```



In [12]:
```
city_sales = df.groupby('City')['Sales'].sum().sort_values(ascending=False)

city_sales.plot(kind='bar')
plt.title("Customer Sales by City")
plt.xticks(rotation=45)
plt.show()
```



In [13]:
```
df['Customer_Type'] = df['Sales'].apply(lambda x: 'High' if x > 30000 else 'Low'

customer_type = df.groupby('Customer_Type')['Sales'].count()

customer_type.plot(kind='bar')
plt.title("High vs Low Value Customers")
plt.show()
```



In [14]:
```
top_customers = df.groupby('Customer_ID')['Sales'].sum().sort_values(ascending=F

print(top_customers)
```

```
Customer_ID
106    110000
101     55000
104     55000
110     55000
102     40000
Name: Sales, dtype: int64
```

Key Insights

- Customers in the 26–35 age group contribute the highest share of total sales, indicating strong purchasing power in this segment.
- Male customers generate more revenue compared to female customers, suggesting higher spending behavior in this group.
- Delhi emerges as the top-performing city in terms of customer sales, indicating a strong market presence and demand in this region.
- A significant portion of revenue is driven by a specific customer segment (age 26–35, male), highlighting a concentrated customer base.

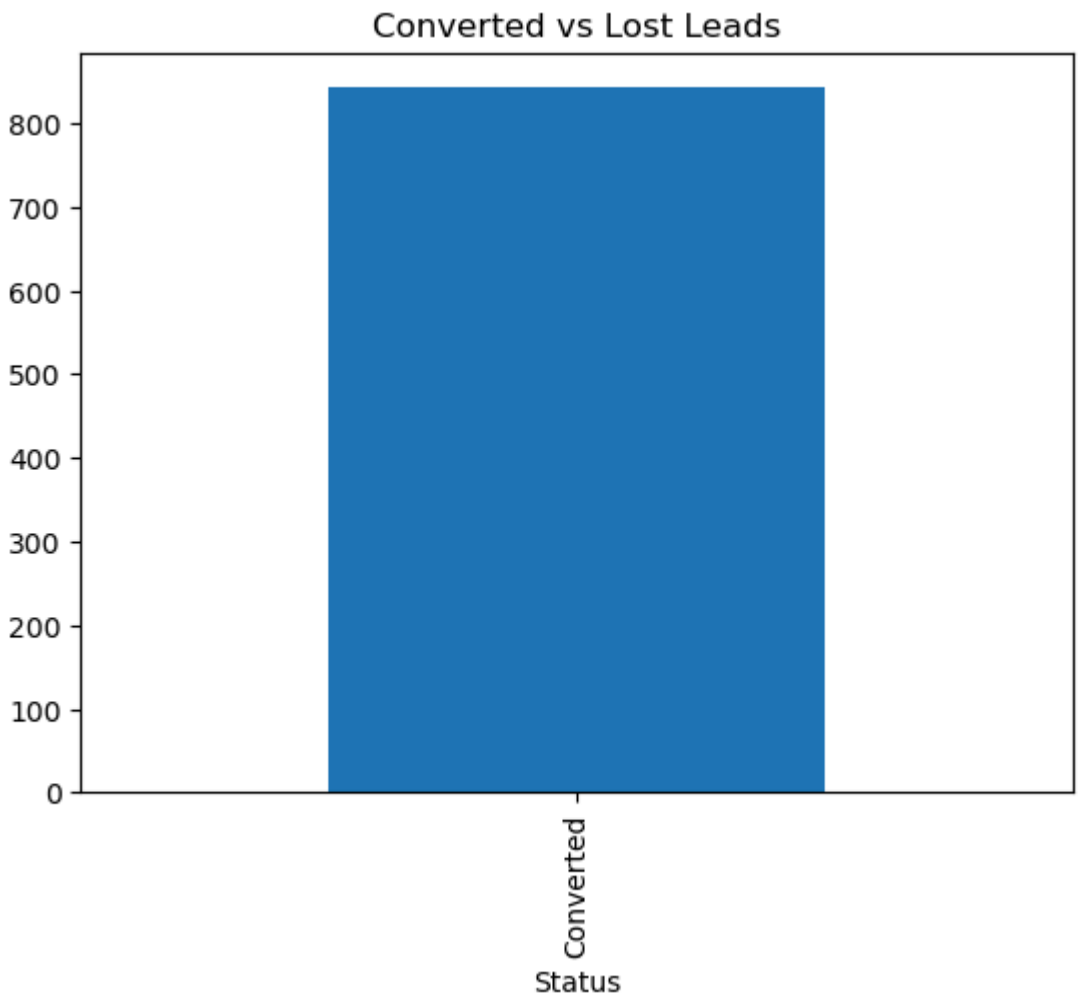
TASK-3 Customer Retention / Churn

In [15]:
```
df = pd.read_csv("D:\CHURN DATA.csv")
print(df.columns)
```

```
Index(['mql_id', 'seller_id', 'sdr_id', 'sr_id', 'won_date',
       'business_segment', 'lead_type', 'lead_behaviour_profile',
       'has_company', 'has_gtin', 'average_stock', 'business_type',
       'declared_product_catalog_size', 'declared_monthly_revenue'],
      dtype='object')
```

In [25]:
```
df['Status'] = df['won_date'].apply(lambda x: 'Converted' if pd.notnull(x) else
```

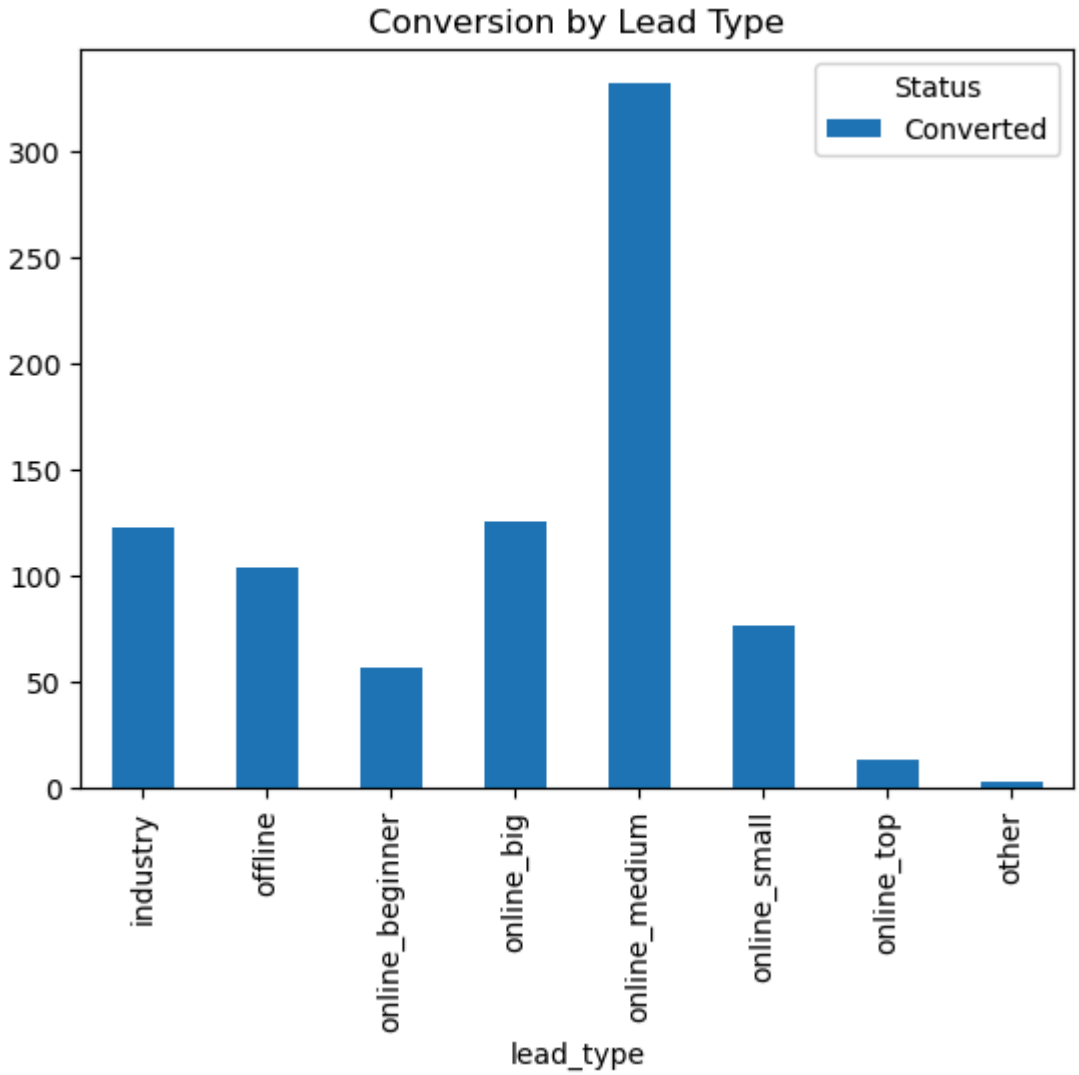
In [26]:
```
df['Status'].value_counts().plot(kind='bar')
plt.title("Converted vs Lost Leads")
plt.show()
```



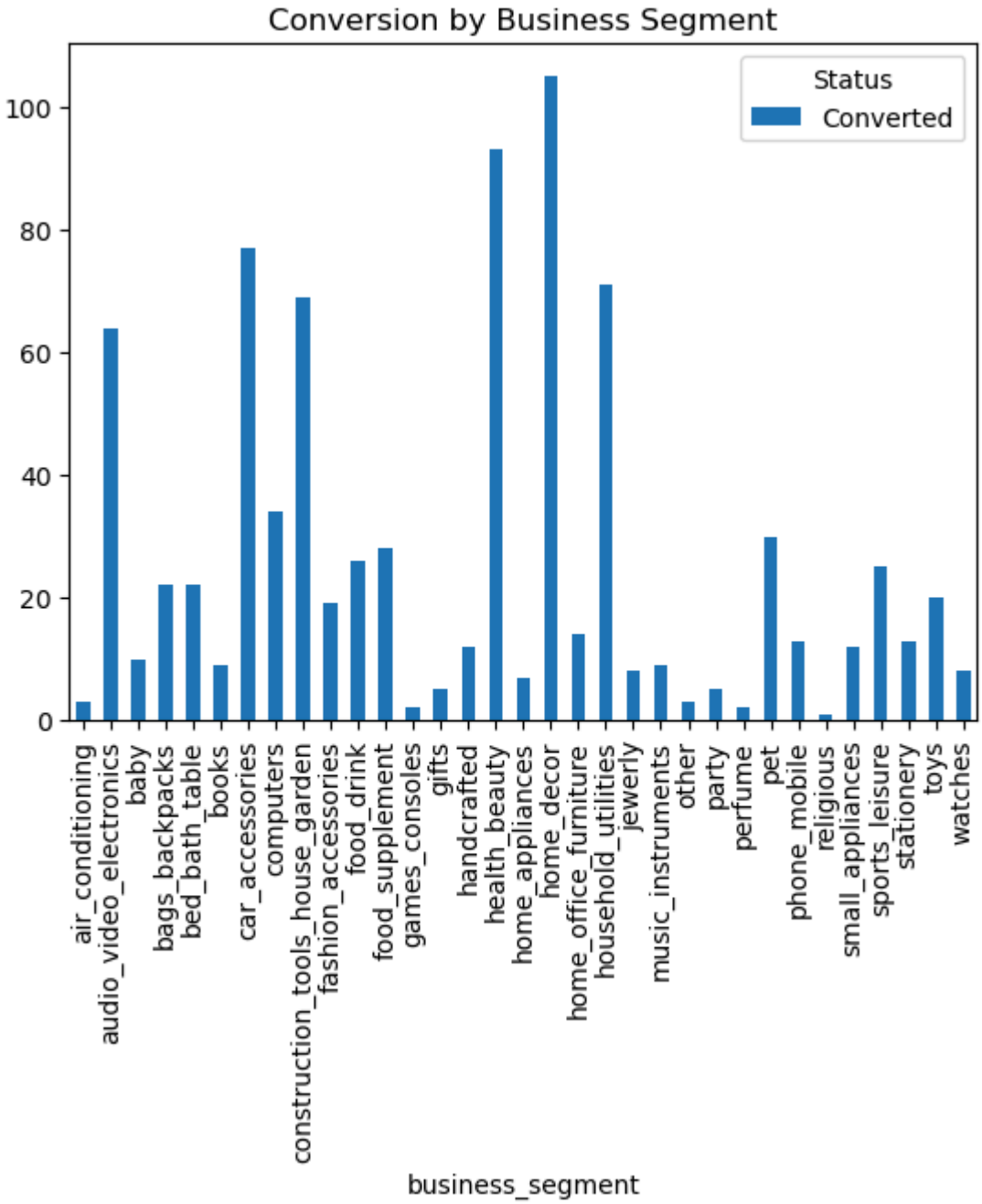
In [27]:
```
funnel_counts = df['Status'].value_counts()
print(funnel_counts)
```

```
Status
Converted    842
Name: count, dtype: int64
```

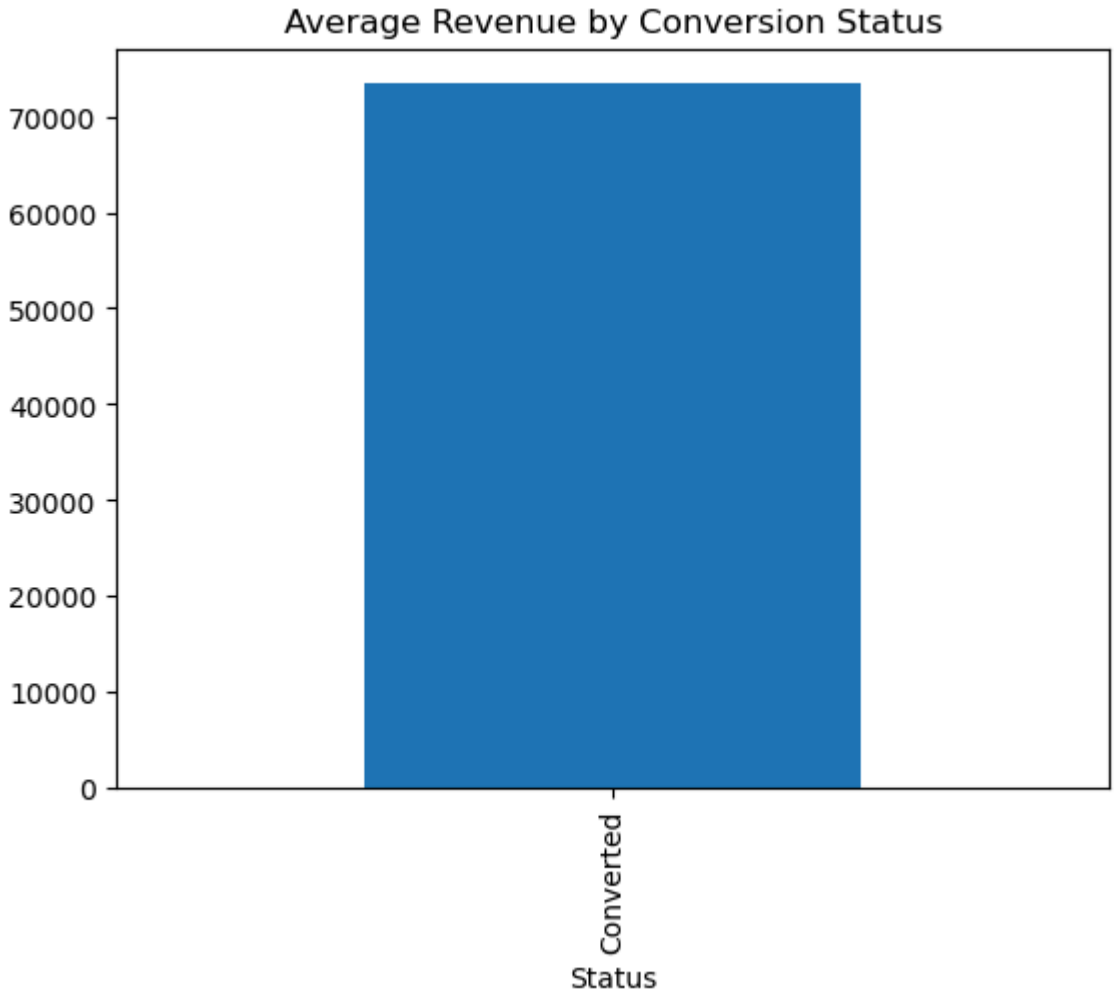
In [28]:
```
df.groupby('lead_type')['Status'].value_counts().unstack().plot(kind='bar')
plt.title("Conversion by Lead Type")
plt.show()
```



In [29]:
```
df.groupby('business_segment')['Status'].value_counts().unstack().plot(kind='bar
plt.title("Conversion by Business Segment")
plt.show()
```



In [30]:
```
df.groupby('Status')['declared_monthly_revenue'].mean().plot(kind='bar')
plt.title("Average Revenue by Conversion Status")
plt.show()
```



FINAL INSIGHTS

- A significant portion of leads are not converted, indicating drop-offs in the sales funnel
- Certain lead types perform better, showing higher conversion rates
- Some business segments have stronger conversion performance than others
- Converted customers tend to have higher average monthly revenue, indicating better quality leads
- There is a clear opportunity to optimize underperforming segments and improve overall conversion rate

In [ ]: